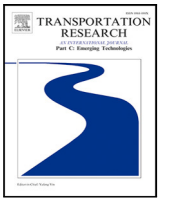
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High-speed rail impacts on intercity accessibility: A multi-modal, multi-scalar networking approach ☆

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ABSTRACT

This paper builds a multi-modal, multi-scalar intercity accessibility framework that integrates multiple layers of transport networks, coordinates actual route/mode choice preferences with the theoretical optimal route planning strategy, and performs accessibility evaluations from a macroscopic planning perspective. The framework is applied to the Guanzhong Plain Urban Agglomeration as a proof-of-concept, particularly focusing on the impact of high-speed railway (HSR) lines, to assess its practical applicability. When compared with the traditional door-to-door approach, our findings indicate that although the spatial distributions of accessibility and its impact of the HSR network are similar to those of the traditional method, our multi-modal, multi-scalar approach enables more precise network configurations that could effectively captures the competition among alternative transport options and the complementary roles of various modes as connectors to intercity transport hubs. Dependence on traditional methods may lead to biased investment evaluations, as our approach demonstrates that the effects of networking methods are intensified in before-and-after analyses that compare changes rather than just evaluating static accessibility.

1. Introduction

China's high-speed rail (HSR) network has experienced swift expansion, reaching 44,500 kilometers of operational lines by the end of 2023 (National Railway Administration of China, 2022). This growth is part of an integrated multi-modal network planning approach, emphasizing improved connectivity and transfer efficiency among various transport methods. This strategy particularly focuses on enhancing links within and between cities, ensuring a more cohesive and efficient transport infrastructure.

Accessibility quantifies the ease for travelers to reach their valued destinations via the transport networks (Levine and Garb, 2002; Levinson and Wu, 2020; Bhat et al., 2000). It counts on the land use development (quality, quantity, and spatial distribution of opportunities) and the mobility on transport network (time, costs, and other effort) from a location's perspective (Cui and Levinson, 2020a; Geurs and Van Wee, 2004), and reflects the interaction strength among these elements (Wu and Levinson, 2020). An increased interest can be found in the application of accessibility measures for project and investment appraisals, as it allows guiding the

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transport and land use planning decisions jointly (Páez et al., 2012; Levine et al., 2017; Anderson et al., 2013; Sun et al., 2023; Li and Wang, 2024; Allen and Farber, 2020).

Attention to the accessibility impacts of the HSR have arisen since the rapid extension of the HSR network around the world. Key issues include the extent to which HSR construction improves accessibility by reducing travel times, as well as equity concerns regarding who specifically benefits from these improvements. For instance, Gutiérrez et al. (1996) evaluates the accessibility impact of the future European high-speed train network based on a weighted average distance. The study reveals that HSR does significantly increase overall accessibility, but it may also worsen disparities between major cities along the HSR network and their surrounding hinterlands. Subsequent studies further support these findings (Cascetta et al., 2020; Bruzzone et al., 2023; Yu and Fan, 2018).

Exceptions do exist, for example, Kim and Sultana (2015) demonstrates that the initial stage of a national HSR network development can intensify spatial imbalances, although efficiency is often prioritized over spatial equity during this period. However, as the network expands, equitable accessibility tends to improve. In a more localized study, Luo and Zhao (2021) finds that HSR slightly mitigates spatial disparities in accessibility within a single province. Additional research has explored the accessibility impacts of HSR across various geographical scales, which reveals that the effects vary depending on the specific planning level (Gutiérrez, 2001; Ortega et al., 2012; Huang and Zong, 2021).

While these studies emphasize accessibility as a valuable tool for guiding policies on HSR operation adjustments and future HSR line planning (Shaw et al., 2014; Wang et al., 2020), they typically focus exclusively on the rail network when modeling intercity travel. This scope fails to capture the connection efficiency between the railway network and other transport modes and ignores the competitive and cooperative dynamics between HSR and other intercity transport options. Few research highlights the importance of 'first and last mile' connectivity at HSR stations and integrates road networks with railways, though many allow driving as the sole access and egress mode (Luo and Zhao, 2021; Wang et al., 2016; Kim and Sultana, 2015). Some studies further explore the combined scenario of driving and HSR for intercity travel, accounting for competition by selecting the shorter travel time as the impedance in accessibility calculations. However, these analyses often rely on the strict assumption that the closest stations to the origin and destination are on the optimal route for the HSR scenario (Yu and Fan, 2018; Luo and Zhao, 2021; Shen et al., 2017).

For the first time, this study develops an intercity accessibility framework using a multi-modal, multi-scalar networking approach, which integrates multiple intercity transport networks, including highways, expressways, general railways (GRs), and HSRs,[1] with the intra-city networks, e.g., city roads, buses, and urban rail transit. This framework allows for feasible transfers between any available modes for each Origin–Destination (OD) pair and captures more precise travel impedance. Such enhanced network modeling and refined cost impedance assessments could improve the practical value of accessibility in integrated network planning (Salonen and Toivonen, 2013).

The framework is applied as a proof-of-concept evaluation for the Guanzhong Plain Urban Agglomeration to verify its applicability. We employ the cumulative opportunity measure to calculate the accessibility to population. Specifically, two scenarios are defined: One considers the complete multi-modal network, while the other excludes the HSR. The spatial distribution features of intercity accessibility and its changes resulting from HSR construction are analyzed comparatively through data visualization. The results can be used to evaluate the current gains and losses associated with HSR construction and also serve as a baseline for assessing future investments, particularly as part of an integrated multi-modal network.

The remainder of the paper is structured as follows: Section 2 provides a comprehensive review of previous studies, focusing on the concept of accessibility and the methods and perspectives used to measure intercity accessibility. Section 3 introduces the study area and summarizes the data used in this study. Section 4 details the multi-modal intercity accessibility framework, explaining how we define travel impedance, construct the OD impedance matrix, and compute accessibility. The results of the multi-modal intercity route planning and accessibility evaluations with respect to the HSR construction are discussed in Section 5. This section also includes a comparison of our multi-modal, multi-scalar networking approach with the traditional door-to-door approach in accessibility evaluations. Section 6 concludes this study.

2. Literature review

2.1. Concept of accessibility

Accessibility was initially defined as "the potential of opportunities for interaction" (Hansen, 1959). The subsequent development of accessibility measures came from various disciplines, addressing problems with different objectives (Wu and Levinson, 2020). A comprehensive review by Geurs and Van Wee (2004) identified four key components in measuring accessibility: transportation, land use, temporal, and individual. Location-based measures, which examine the interaction efficiency between transportation and land use components, are typically used as performance measures to support decision-making in urban planning and transport investment projects (Levine et al., 2017; Anderson et al., 2013; Cui et al., 2017).

As illustrated in Fig. 1, the transport component in location-based measures reflects the configuration (e.g., network layout and structure) and efficiency (e.g., speed) of the transport system, and it is measured by the travel impedance, such as time and expenses, between origins and destinations (Levinson and Wu, 2020; Cui et al., 2024; Lucas, 2012; Da Silva et al., 2022). The land use component expresses the location and quality of available opportunities that target specific travel purposes in intra-city scenarios,

[1] Some may argue that HSR and GR are part of the same railway system. However, we consider them as distinct modes because, in the context of China, they typically operate through separate stations, and many previous studies on intercity accessibility have not integrated the GR with the HSR network.

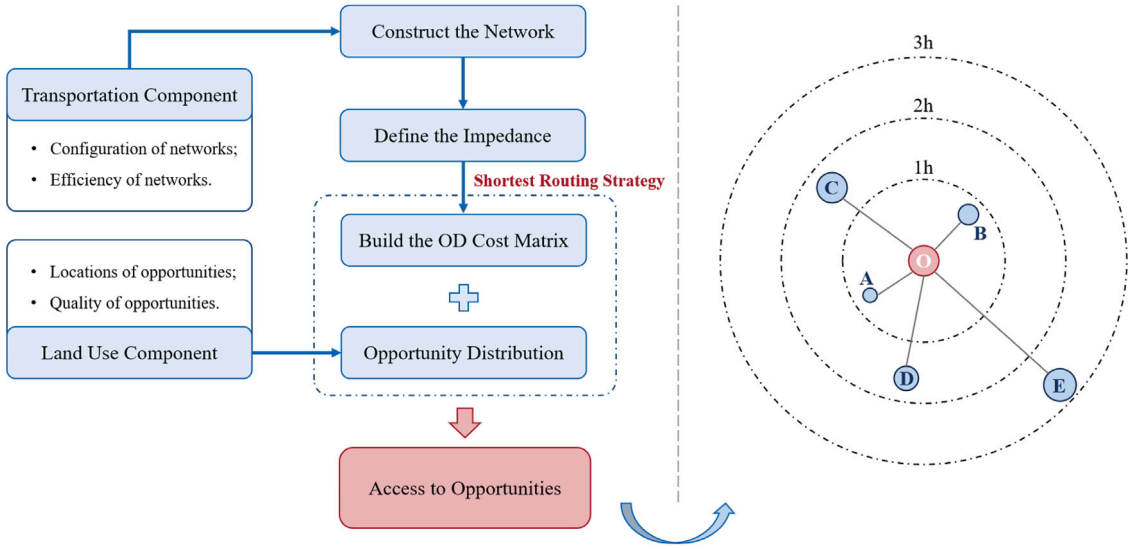


Fig. 1. Concept of location-based accessibility: The red dot represents the origin location, while the blue dots represent valued destinations, with their size indicating the quality of available opportunities. The circular areas around the dots illustrate travel time isochrones from the origin.

like jobs (Bastiaanssen et al., 2022; Hu, 2017), resident workers (Levinson et al., 2017; Li et al., 2021), education (Pizzol et al., 2021; Sharma and Patil, 2022), or health care (Guagliardo, 2004; Mao and Nekorchuk, 2013; Gong et al., 2021). For inter-city accessibility evaluations, land use generally considers broader socioeconomic indicators, such as GDP (Gutiérrez, 2001), population (Zhou et al., 2018), or mass of destination that combines both aspects (Luo and Zhao, 2021; Jiang et al., 2023).

The calculation procedures for various location-based accessibility measures remain consistent, encompassing steps of defining travel impedance, constructing the origin–destination (OD) impedance matrix, and computing accessibility.

- **Define the travel impedance:** What types of cost components are taken into account, and how are they incorporated?
  Research on accessibility focuses more on travel time (Luo and Zhao, 2021; Monzón et al., 2013; Cavallaro et al., 2023; Luo et al., 2023; Chen et al., 2020), but the methods for measuring it differ significantly. For intra-city travel, some studies consider only the in-train travel time as the impedance for accessibility calculations (Cao et al., 2013; Huang and Zong, 2021; Jiang and Chu, 2017), while others also include the transfer times required between different railway lines (Moyano et al., 2018a; Liu et al., 2021a; Shaw et al., 2014). The door-to-door approach additionally accounts for the time needed to access to and egress from railway stations, reflecting the total travel time of the journey (Salonen and Toivonen, 2013).
  However, some researchers argue that an exclusive focus on travel time may lead to biased assessments of accessibility that overstate the advantages of faster but more costly routes or modes (Cui and Levinson, 2018). Only a few studies have examined the significance of out-of-pocket monetary costs incurred during travel (Cascetta et al., 2020; Avogadro et al., 2021; Vale, 2020).
- **Build the OD impedance matrix:** How much travel cost is needed to travel from each origin to each destination?
  Network Analysis and Cost Distance Modules of ArcGIS are widely used to build the OD impedance matrix for accessibility calculations, as summarized in Table 1. These modules fundamentally rely on Dijkstra's algorithm (ESRI, 2021), which find the shortest route based on the specified impedance from all origins to all destinations. This approach embodies the core idea of location-based accessibility, creating 'ideal' conditions to measure a location's potential capability to access valuable destinations through the transportation system. Actual reachable opportunities for specific individuals may be lower (Dong et al., 2006), a concern more thoroughly addressed in person-based (Miller, 1991; Kwan, 1998) and utility-based accessibility studies (Handy and Niemeier, 1997; Ben-Akiva, 1985) that incorporate both temporal and individual components of accessibility.
  However, existing GIS platform modules may have limitations in complex network modeling scenarios, such as inter-network transfers. Therefore, some studies have developed their own networks to construct the cost matrix independently (Shaw et al., 2014; Cao et al., 2013), while traditional Dijkstra's algorithm continues to be used as the primary tool for shortest path searching in these analyses.
- **Compute accessibility:** How are travel impedance and opportunities addressed?
  Wu and Levinson (2020) provides an overview of accessibility measures based on their approach to handling travel impedance and opportunities. Among them, cumulative opportunity measure (also referred to as contour measures or isochronic measures) stands out as the most frequently referenced 'default' measure in the literature (Wachs and Kumagai, 1973). This measure applies a rectangular impedance function that delineates an isochrone (e.g., 1-hour or 2-hour driving radius) from each origin

Table 1
Representative studies on inter-city accessibility studies.

Topics	Authors	Network Modeling					Accessibility Calculation		
		Access	→	In-train	→	Egress	Path Search	OD Cost Matrix	Measures
Access and Egress	Diao et al. (2017)	×	√	–	√	×	–	C_{Auto}, C_{PT}	WATT
	Moyano et al. (2018b)	×	√	–	√	×	ArcGIS	C_{Auto}, C_{PT}	WATT
	Wu and Han (2022)	×	√	–	√	×	–	C_{Auto}, C_{PT}	WAD
	Xu et al. (2023)	×	√	–	√	×	–	C_{Auto}, C_{PT}	WATT, CUM, PA
City-to-City	Cao et al. (2013)	–	–	√	–	–	Dijkstra	C_{HSR}, C_{GR}, C_{Air}	WATT, CUM, PA
	Shaw et al. (2014)	–	–	√	–	–	Dijkstra	C_{HSR+GR}	ATT
	Jiang and Chu (2017)	–	–	√	–	–	–	C_{HSR+GR}	WATT, PA
	Moyano et al. (2018a)	–	–	√	–	–	–	C_{HSR}	PA
	Liu et al. (2021b)	–	–	×	–	–	–	C_{HSR}, C_{GR}	Improved PA
	Huang and Zong (2021)	–	–	√	–	–	ArcGIS	$C_{HSR+GR+Auto+Water}$	WATT, PA
Door-to-Door	Ortega et al. (2012)	×	√	∅	√	×	ArcInfo GIS	$C_{HSR+Auto}$	PA
	Kim and Sultana (2015)	×	√	∅	√	×	ArcGIS	$C_{HSR+Auto}$	WATT, PA
	Wang and Duan (2018)	∅	√	√	√	∅	ArcGIS	$C_{Intra-city+Inter-city}$	TT, WATT
	Yu and Fan (2018)	×	√	∅	√	×	ArcGIS	$\min(C_{Auto}, C_{HSR})$	WATT, PA
	Shen et al. (2017)	∅	√	√	√	∅	–	$\min(C_{Auto}, C_{HSR+GR})$	WATT
	Luo and Zhao (2021)	×	√	∅	√	×	–	$\min(C_{Auto}, C_{HSR+Auto})$	PA
	Jiang et al. (2023)	×	√	×	√	×	–	C_{HSR}, C_{Air}	CUM, PA
	This Study	√	√	√	√	√	Rule-constrained optimal route planning	$C_{HSR+GR+Auto+PT}$	CUM

1 The arrow (→) represents transfers between intra-city and inter-city segments. ∅ denotes the case where the papers do not specify whether transfers are allowed.
2 C_{mode} denotes the travel cost between origins and destinations, in which 'mode' refers to any modes of transportation. Notably, the symbol '+' signifies the integration between different modes, indicating that transfers are allowed in between.
3 TT: travel time; ATT: average travel time; WATT: weighted average travel time; WAD: weighted average distance; CUM: cumulative opportunity measure; PA: potential accessibility; PT: public transit; GR: general railway; HSR: high-speed railway.

point, and then counts all opportunities within the boundary.[2] Negative exponential functions are also widely used, well known as gravity-based measures or potential accessibility measures (Ingram, 1971; Vickerman, 1974). They also quantify the number of accessible opportunities but assign greater weight to those that are closer rather than farther away.[3] These measures are widely used in intercity accessibility studies, as shown in Table 1.

2.2. Intercity accessibility

Recent studies have extensively examined the accessibility in the context of intercity travel, particularly focusing on the assessment of the HSR network. Based on their modeling scenarios, we have classified these studies into three categories: access and egress, city-to-city, and door-to-door, as illustrated in Table 1.

Access and egress studies emphasize the 'first and last mile' of intercity travel, examining accessibility to intercity transport nodes via both auto and public transit. These studies are particularly relevant for evaluating the location of stations and their efficiency in integrating with urban transport networks (Brons et al., 2009; Givoni and Rietveld, 2007).

In contrast, the city-to-city category focuses on the intercity segment, treating each city as a point when modeling the network. It aims to assess the quality of intercity connections. Some studies within this category consider network integration when modeling the network. For instance, Jiang and Chu (2017) evaluated the accessibility benefits and potential equity issues associated with HSR construction and policy development by integrating the GR network with HSR. Huang and Zong (2021) expand it to further include roadways and water transport alongside HSR and GR. However, both studies do not account for penalties associated with transfers when calculating city-to-city travel times, potentially leading to an overestimation of accessibility benefits.

The door-to-door approach, as proposed by Salonen and Toivonen (2013), treats the city as an area rather than a single point, integrating access and egress with the city-to-city segments. This method offers a more accurate estimation of intercity travel time, by considering all journey segments. However, many studies focus solely on auto travel as the connection to railway stations, neglecting the important role of public transit in urban areas. Particularly, some developed cities, like Xi'an and Xianyang, feature well-developed metro and bus services that support intercity transport hubs. Furthermore, the routing strategies used to construct the OD cost matrix in these studies tend to be overly simplistic, e.g. overlooking potential transfers in each segment, as summarized in Table 1, or assuming auto travel to the nearest railway station when determining the shortest path (Ortega et al., 2012; Luo and Zhao, 2021; Yu and Fan, 2018).

[2] In the diagram shown in Fig. 1, the cumulative opportunity measure assesses accessibility from origin O within a 1-hour time threshold as the sum of opportunities at destinations A and B.

[3] In the diagram shown in Fig. 1, potential accessibility evaluates access from origin O as the sum of opportunities across all destinations, with each destination assigned a weight based on a negative exponential function. The weights for destinations A and B are greater than those for C and D, which in turn are greater than for E.

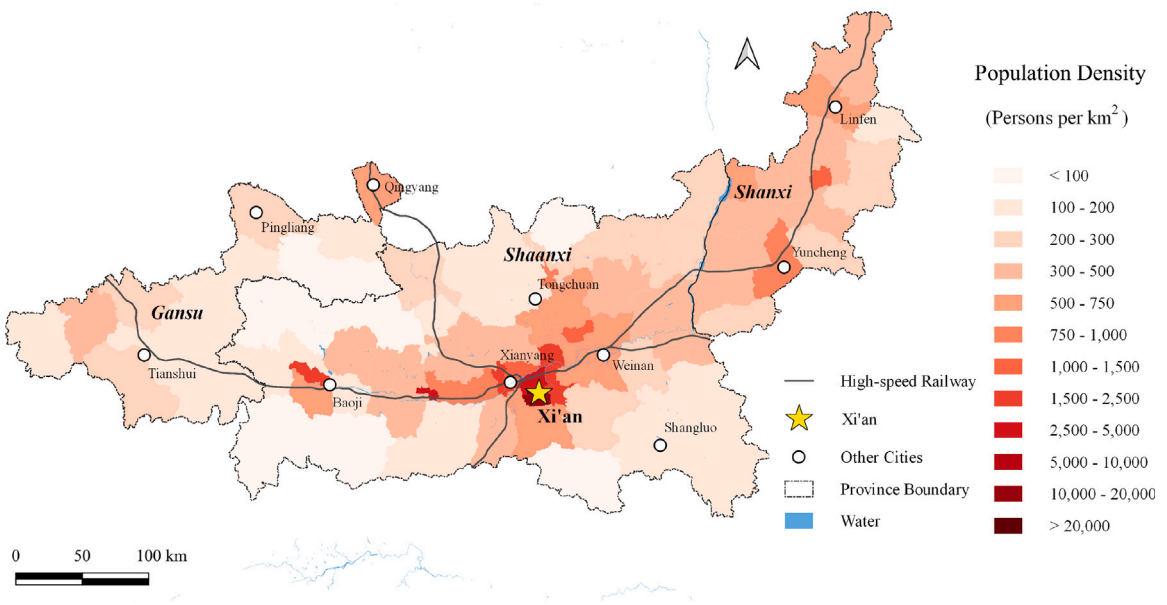


Fig. 2. The study area: Guanzhong Plain Urban Agglomeration.

3. Data collection

This paper studies the Guanzhong Plain Urban Agglomeration, the ninth officially approved national-level urban agglomeration in China. The region centers on Xi'an, an historic capital of China and one of China's central cities, and includes another 10 cities located in Shaanxi, Shanxi, and Gansu provinces, see Fig. 2. It covers a total area of 107,100 km^2, and has built an integrated land transport network, consisting of highways, expressways, GRs, and HSRs, as for intercity travel, as well as city roads, buses, and urban rail transit, as for intra-city travel. Note that the aviation network is disregarded in this study, as the air traffic demand inside of the Guanzhong Plain Urban Agglomeration is quite limited due to a short travel distance in a relative term. Also, no direct flights are available within the study area.

Geographical boundaries, travel impedance by mode, economic and population data, and migration data are collected to measure the accessibility. Details are discussed as follows.

- **Geographical boundaries**
  The geographical boundary of each county in the Guanzhong Plain Urban Agglomeration is extracted from the National Basic Geographic Databases provided by the National Catalogue Service for Geographic Information (National Geomatics Center of China, 2023), and calibrated based on the online base map as well as the official administrative division defined by the Ministry of Civil Affairs. In all, the study area encompasses 90 counties and their centroids are selected as the origins and destinations for accessibility calculations.
- **Travel impedance**
  The measure of composite intercity travel accessibility requires travel impedance data between vital nodes on the multi-modal network, including origins, destinations, and intercity transportation hubs, e.g., HSR or GR stations. Travel time and monetary cost by auto, by transit, by HSR, and by GR are collected via multiple data sources.

  – **By transit and by auto**
  Transit and auto travel data from origins to destinations (for the cases that the route by a single mode, either transit or auto, is the optimal solution) and from origins/destinations to intercity transportation hubs (for the cases that the optimal path requires transfers among different modes, e.g., transit – HSR – transit),[4] are collected from the Amap Open Platform via the Route Planning Web API (Amap, 2023). This module selects the optimal route for an OD pair by a given mode based on a predetermined routing strategy and outputs the corresponding travel impedance records, including travel time, travel distance, and travel cost, like fares for transit and tolls for auto (Yu and Cui, 2023). We extract the data of the shortest time path for further analysis in this study. Note that valid routes by auto are found for all these

[4] We do not consider non-motorized vehicles or walking as suitable connecting modes for directly reaching intercity transport hubs. This exclusion is based on the fact that these modes are generally not luggage-friendly, and the connecting distances to transport hubs are often too long to be practical for walking. However, we assume people typically walk short distances with luggage to access public transport services, taxis, or cars.

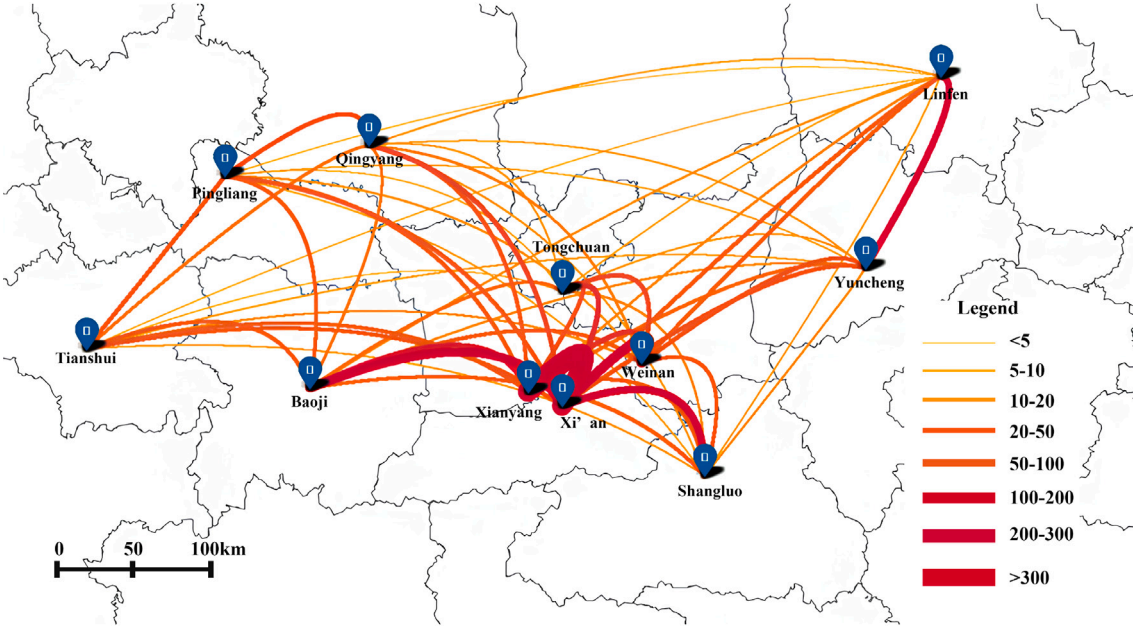


Fig. 3. The migration intensity between any of the two cities in the study area.

predetermined OD pairs, but there are some failed queries when we search the transit routes, because the coverage of the urban transit networks are bounded. In this case, transit is considered as an invalid mode for these corresponding connections.

– **By HSR and by GR**
Travel time and ticket price between any two rail stations, including both HSR and GR stations, in the study area are obtained from the China Railway Official Website. Note that transfers inside of the railway system, e.g., HSR - HSR, GR - GR, HSR - GR, are all allowed when searching for the optimal path, so we record both direct route and transfer route plans offered by the website.

- **Economic and population data**
Population by county and its average annual disposable income are collected from the 2022 Statistical Year Book of Shaanxi (Statistics Bureau of Shaanxi Province, 2022), Shanxi (Statistics Bureau of Shanxi Province, 2022), and Gansu (Statistics Bureau of Gansu Province, 2022) province (updates for 2023 have not been released yet). Population is regarded as the relevant opportunity in the cumulative accessibility measure. The average annual disposable income is used to evaluate the value of travel time (VoTT), which is further applied to combine the travel time and monetary cost en-route, working as the travel impedance when calculating accessibility. The detailed method can be found in Section 4.3.
- **Migration data**
Baidu Map, a popular map service provider in China, provides the migration scale index data (Baidu, 2023), which tracks the intensity of interaction between any two cities among the 367 major cities in China based on the emigration and immigration counts correspondingly captured by billions of Location Based Services (LBS) data (Wei and Wang, 2020; You et al., 2023), as shown in Fig. 3. A higher value of the migration scale index on an origin–destination level refers to a greater extent of population movement from one city to another. Note that the term 'migration data' used by Baidu does not refer to permanent changes in residence location, but instead to intercity travel flow from one city to another. We acquire the 2022 daily migration scale index data, which are used for the calibrations of the cost decay function particularly for intercity travel. The results are further applied to calculate the cost-weighted access for a comprehensive evaluation of the HSR impacts.

4. Multi-modal intercity accessibility framework

4.1. Defining the travel impedance

This study considers both time and monetary costs to determine the optimal mode(s) and routes for a specific OD pair within the multi-modal, multi-scalar network. These costs constitute the primary private cost items, accounting for over 85% of the total (Cui et al., 2024). Other factors such as comfort (e.g., seating and personal space) (Bouscasse and de Lapparent, 2019), convenience (e.g., the number of transfers) (Algers et al., 1975; Li et al., 2018), familiarity (e.g., prior knowledge or experience) (Lu et al., 2020; Lotan, 1997), and travel time reliability (e.g., the risk of arriving late) (Zhu et al., 2021; Moghaddam et al., 2019; Liu et al.,

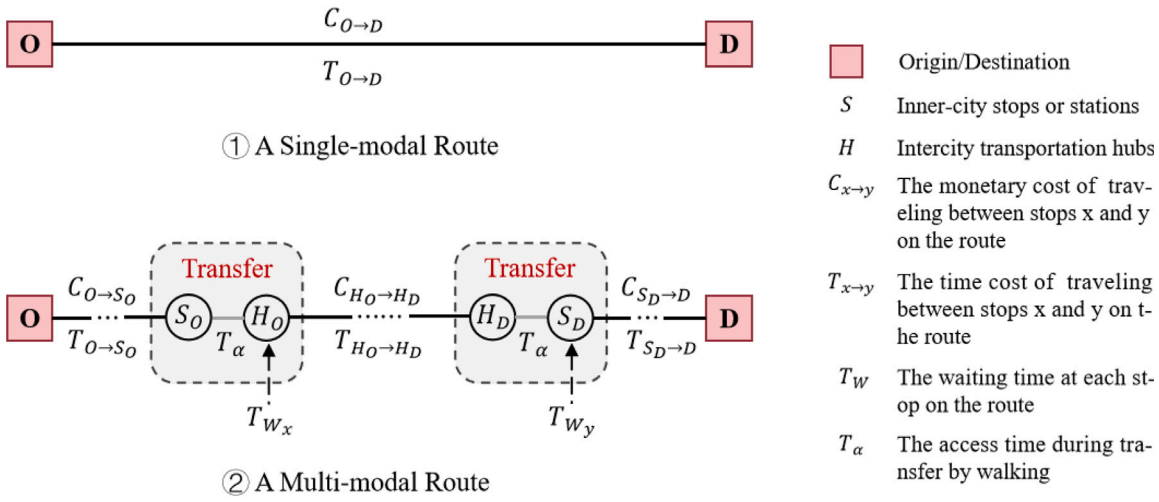


Fig. 4. Illustration of travel cost en-route for single-modal and multi-modal, multi-scalar routes. Note that the diagram highlights inter-modal transfers between inner-city stops or stations (S) and intercity transportation hubs (H). Transfers occurring within inner-city or inter-city segments are simplified and depicted by ellipses.

2004), may also affect route choices. We acknowledge that individual accessibility may be influenced by these factors to varying degrees, depending on personal preferences. However, our analysis does not model these other factors from a macroscopic planning perspective (Cui and Levinson, 2021; Schwanen and Lucas, 2011). Recent studies have increasingly considered time and monetary costs together in evaluations of location-based accessibility (Vale, 2020; El-Geneidy et al., 2016; Yu and Cui, 2023), which further supports the validity of this approach in our analysis.

There are two possible scenarios of route composition: single-modal route and multi-modal, multi-scalar route, as illustrated in Fig. 4.

- **Single-modal route:** This specifically refers to trips where travelers use only a single mode of transport, such as driving or taking transit, from the origin to the destination without any transfers between different networks.[5] In this scenario, the time cost is equivalent to the in-vehicle travel time for the entire trip. Similarly, the monetary cost includes the expenses for fuel and tolls for driving, or fares for public transit, throughout the journey.

$$T_{OD} = T_{O\to D} \tag{1}$$
$$C_{OD} = C_{O\to D} \tag{2}$$

where:
T_{OD} and C_{OD}: total travel time and monetary costs from origin O to destination D, respectively.;
$T_{O\to D}$ and $C_{O\to D}$: travel time and monetary costs for the specific trip segment from O to D, respectively.

- **Multi-modal, multi-scalar route:** This scenario involves trips where travelers use multiple modes of transport to reach their destination, thus transfers are needed along the way. Typically, the route is segmented into three parts: intracity travel within the origin city, intercity travel between the origin and destination cities, and intracity travel within the destination city. At least two transfers are required, one at each junction between the intracity and intercity segments. Additional transfers within these segments, such as between bus transit and metro or between GR and HSR, are also permitted.
In this scenario, the time cost encompasses both the in-vehicle/in-train travel time and the waiting time at transfer nodes throughout the entire trip. The monetary cost includes fares for public transport modes and tolls and fuel costs for private vehicles.

$$T_{OD} = \sum_{i=1}^{n_1} T_{O\to S_O}^{\ i} + \sum_{j=1}^{n_2} T_{S_O\to S_D}^{\ j} + \sum_{k=1}^{n_3} T_{D\to S_D}^{\ k} + \sum_{q=1}^{n} (T_{W,q} + T_{\alpha,q}) \tag{3}$$
$$C_{OD} = \sum_{i=1}^{n_1} C_{O\to S_O}^{\ i} + \sum_{j=1}^{n_2} C_{S_O\to S_D}^{\ j} + \sum_{k=1}^{n_3} C_{D\to S_D}^{\ k} \tag{4}$$

where:
S_O and S_D: junctions between the intracity and intercity trip segments on the origin side and destination side, respectively;

[5] Transit trips that include transfers between different buses or between bus and subway are still considered a single-modal trip.

$T_{O\to S_O}$ and $C_{O\to S_O}$: in-vehicle or in-train travel time and monetary costs incurred during the trip segment from O to S_O using a specific mode;
$T_{S_O\to S_D}$ and $C_{S_O\to S_D}$, as well as $T_{S_D\to D}$ and $C_{S_D\to D}$ follow the same definition rules as $T_{O\to S_O}$ and $C_{O\to S_O}$, applying to their respective trip segments;
$T_{W,q}$ and $T_{\alpha,q}$: waiting time and walking access time during the q_{th} transfer;
n_1, n_2, n_3: number of different transport modes used in the trip segments from O to S_O, from S_O to S_D, and from S_D to D, respectively;
n: total number of transfers during the journey.

In the theoretical equation, it is assumed that each transfer incorporates both access time and waiting time in the total travel time for the trip. In practice, access time is measured based on the network distance between two nodes for transfer (e.g., from a metro station to a railway station or from parking to a railway station) and an average travel speed set at 4.5 km/h. For waiting time, practical data extracted from the Amap Open Platform for transit travel does not account for waiting times at bus stops or subway stations. Despite this, the impact is minimal for inter-city travel due to the relatively high frequency of services, with departure intervals ranging from approximately 3 to 5 min in the case of Xi'an (Yu and Cui, 2023). Additionally, a 30-minute waiting time is applied at the origin station (S_O), based on the assumption that travelers arrive at GR and HSR stations 30 min prior to departure to allow for security checks and to locate the correct gate (Luo and Zhao, 2021; Wang et al., 2016; Sánchez-Mateos and Givoni, 2012). Transfer time within the intercity network, including GR–GR, HSR–HSR, and GR–HSR transfers, are determined based on intercity railway timetables, incorporating both access and waiting times. At the station near the destination side (S_D), the waiting time is assumed to be zero.

Monetary costs associated with additional food or hotel expenses during transfers are not considered in this study.

4.2. Building the OD impedance matrix

Building the OD impedance matrix comprises two main steps: first, constructing a multi-modal, multi-scalar network graph, and second, identifying the optimal cross-network route between OD pairs. The travel impedance along the identified routed is then recorded as corresponding elements in the OD matrix. Details of each step are discussed in this subsection.

4.2.1. Constructing a network graph

The multi-modal, multi-scalar network for intercity travel integrates urban roads, bus transit, and metro for intracity connections, along with intercity railways (including GRs and HSRs), intercity buses, and highways for intercity routes. This network is constructed based on the transfer graph approach, as proposed by Ayed et al. (2008), which integrates multiple transport layers through the use of virtual nodes for seamless transfers, as depicted in Fig. 5. These virtual nodes are strategically positioned at transport hubs where various modes are accessible within a walkable distance, marked as stars in the figure. Virtual edges within these nodes, represented by the dash line, enable transfers between networks. The impedance of these virtual edges is quantified based on the walking time required to connect these different modes. A prime example is the Xi'an HSR station, around which a virtual node is defined to facilitate transfers among automobiles, bus transit, three metro lines, and high-speed rail lines.

4.2.2. Specifying the optimal route principle

The principle for selecting the optimal route is to minimize both time and monetary costs en-route. To simplify the multi-objective optimization problem into a single objective, we integrate travel time and monetary cost by converting travel time into a monetary unit based on the average annual disposable income, adjusted for population weights (Yu and Cui, 2023).

$$G = T * \mathrm{VoTT} + C \tag{5}$$

where:
G: generalized travel cost;
VoTT: value of travel time.

The VoTT can be defined in multiple ways to measure the travel impedance between OD pairs, accommodating various analytical needs:

- A fixed VoTT value across the region;
- A VoTT drawn from a known distribution;
- A VoTT drawn from a known distribution for each traveler;
- A VoTT that varies by metropolitan area but remain fixed within each area;
- A VoTT that varies by metropolitan area, drawn from a known distribution for each area;
- A VoTT that varies by metropolitan area, drawn from a known distribution for each traveler within each area.

Further refinement of VoTT could consider variations by mode, by segments of travel, by purpose, and so on, to enhance the granularity and accuracy of the travel impedance assessment.

In our study, we employ a fixed VoTT across the region for our proof-of-concept, considering the following reasons. Primarily, the focus of this study is designed to serve as a planning tool to support decision-making in transport and land use investments. Decision-makers generally prefer definitive conclusions, such as "Proposal A is better than Proposal B", over probabilistic outcomes,

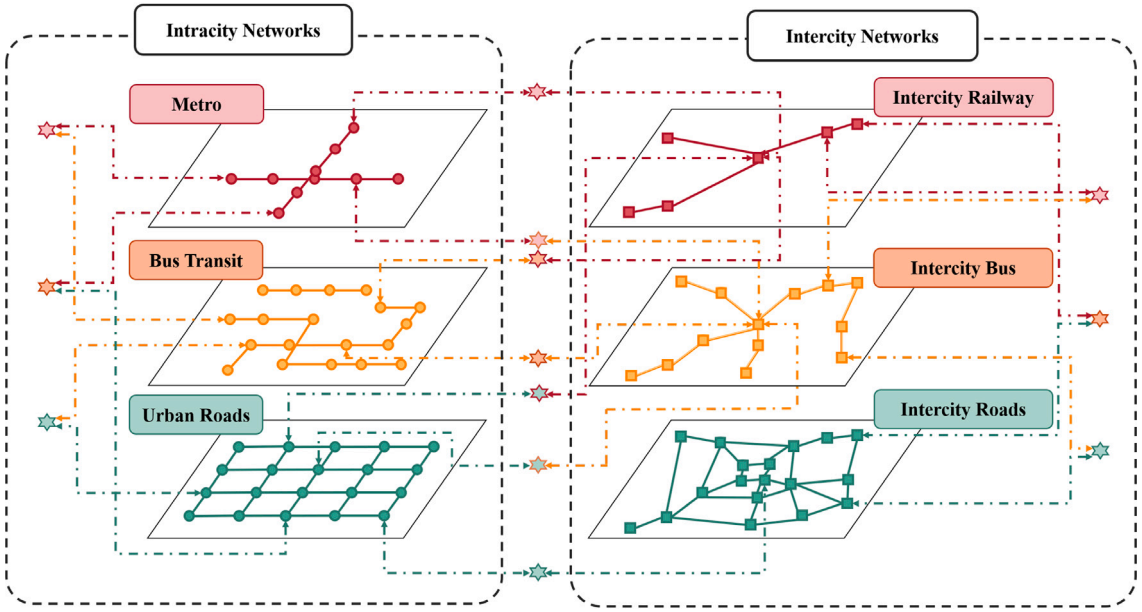


Fig. 5. Illustration of a multi-modal, multi-scalar transport network: Circles represent intracity network nodes, squares denote intercity network nodes, and stars indicate virtual nodes that allow transfers between different network layers.

such as "Proposal A is better than Proposal B with an 80% probability". Furthermore, the lack of detailed disposable income data, particularly at the extensive geographical scale required for intercity travel analysis, due to privacy concerns, limits the accurate determination of VoTT distributions. Hence, in line with previous accessibility studies (Avogadro et al., 2021; Cui and Levinson, 2018; El-Geneidy et al., 2016; Ma et al., 2017; Yu and Cui, 2023), a fixed VoTT is chosen to ensure clarity and consistency in our evaluations.

4.2.3. Defining the rule-constrained multi-modal route planning strategy

Dijkstra's algorithm is commonly used for shortest path search problems and has been embedded in many GIS software for network analysis (Karas and Demir, 2011; ESRI, 2021). However, these methods, which primarily solves single-objective optimization problems, may not always identify feasible routes for cross-network route planning. Instead, it could return solutions with excessively frequent and unrealistic transfers, such as driving to a HSR station after taking the subway, followed by another drive to a bus stop upon alighting from the HSR, as shown in Fig. 6. Furthermore, some routes returned for specific OD pairs can appear even more impractical, especially considering that the railway route set might include transfers between HSR and GR, and the public transit route set allows transfers between city buses and urban railways. Such routing outcomes often differ from common sense expectations for route and mode choices, necessitating an adapted algorithm that better reflects realistic travel behaviors.

Hence, we extend the traditional shortest path searching algorithm by introducing restriction rules that limit the number of transfers between different transport modes for each segment of a multi-modal route and across the entire journey for single-modal routes. This adjustment ensures that the theoretical optimal route planning strategy better corresponds with practical travel behaviors. A detailed description of these rules and the route planning algorithm is available in Appendix A.

Note that this rule-constrained multi-modal route planning strategy can be integrated with various shortest path algorithms. In this study, we chose to use the fundamental Dijkstra's algorithm because its efficiency is adequate for constructing a moderate-sized impedance matrix. For instance, in our case study, it takes around 15 min to build a 90×89 matrix for a complete iteration on a standard computer configuration (12th Gen Intel(R) Core(TM) i7-12700, 2.10 GHz, 16.0 GB memory). Additionally, using Dijkstra's algorithm is consistent with previous accessibility studies, facilitating straightforward comparisons. More sophisticated algorithms, such as the A* Algorithm or Ant Colony Optimization, can be used in future studies addressing larger study areas to improve efficiency and responsiveness.

4.3. Computing accessibility

The cumulative opportunity measure counts the number of reachable opportunities given a predetermined threshold, either time or cost, or any other factors that are concerned as the travel impedance in the evaluations (Wachs and Kumagai, 1973; Vickerman, 1974; Ingram, 1971). It shows advantages on simplicity and interpretability over other sophisticated measures, so has been widely used in planning practice (El-Geneidy and Levinson, 2006).

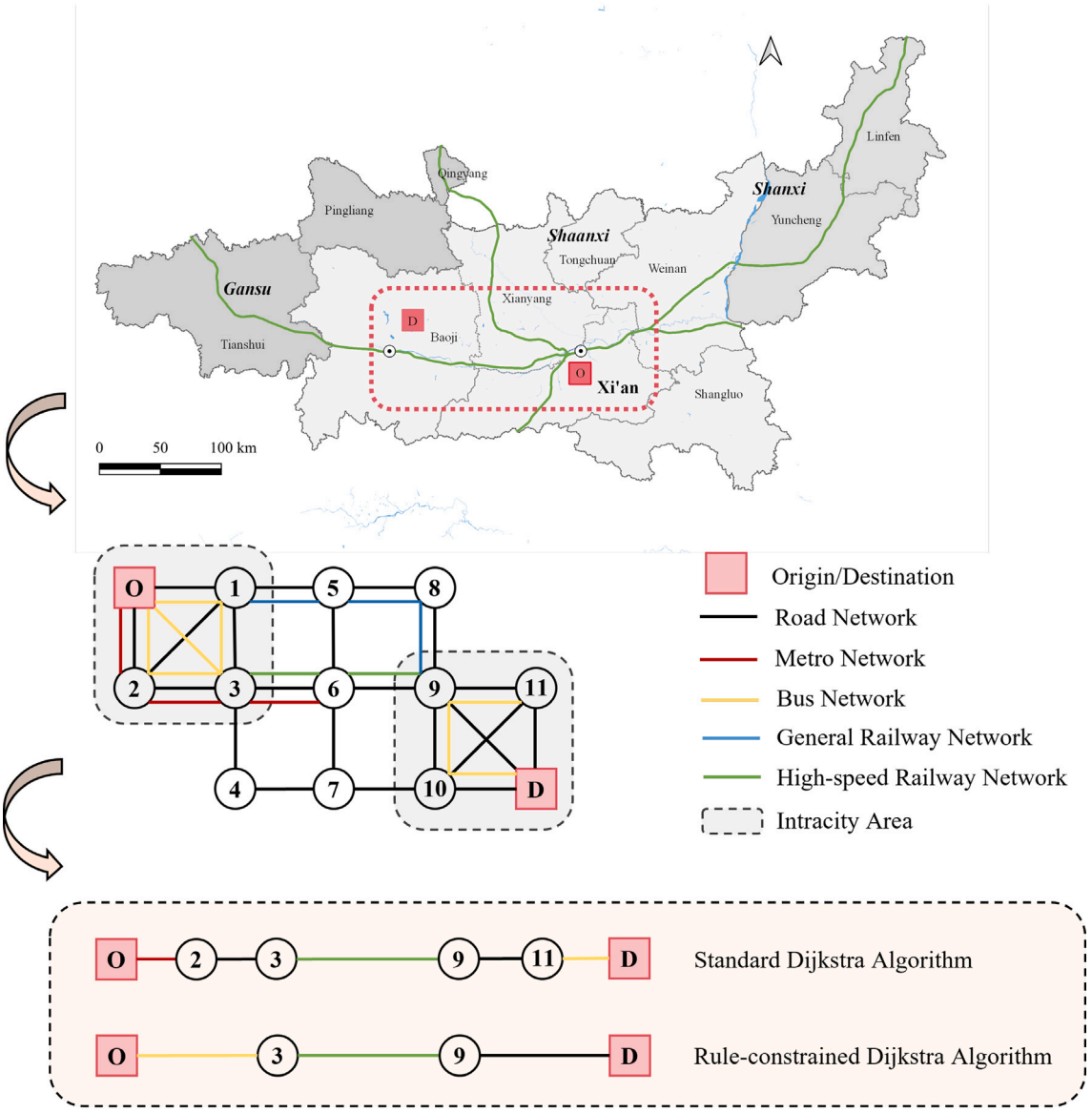


Fig. 6. Example of a multi-modal route from origin O to destination D.

The cumulative opportunity measure can be written as,

$$A_{i,\tau} = \sum_j M_j \cdot f(C_{ij}, \tau) \tag{6}$$

$$f(C_{ij}, \tau) = \begin{cases} 1 & \text{if } C_{ij} < \tau \\ 0 & \text{if } C_{ij} \geq \tau \end{cases} \tag{7}$$

where,

$A_{i,\tau}$: accessibility of county i in the cost threshold τ;

M_j: population at destination j;

C_{ij}: travel impedance between i and j;

τ: cost threshold;

$f(C_{ij}, \tau)$: impedance function.

We calculated the cost-weighted average, which merges the accessibility measurements of different cost thresholds with a decay parameter, following the consensus that opportunities nearby are more valuable than those farther away (Anderson et al.,

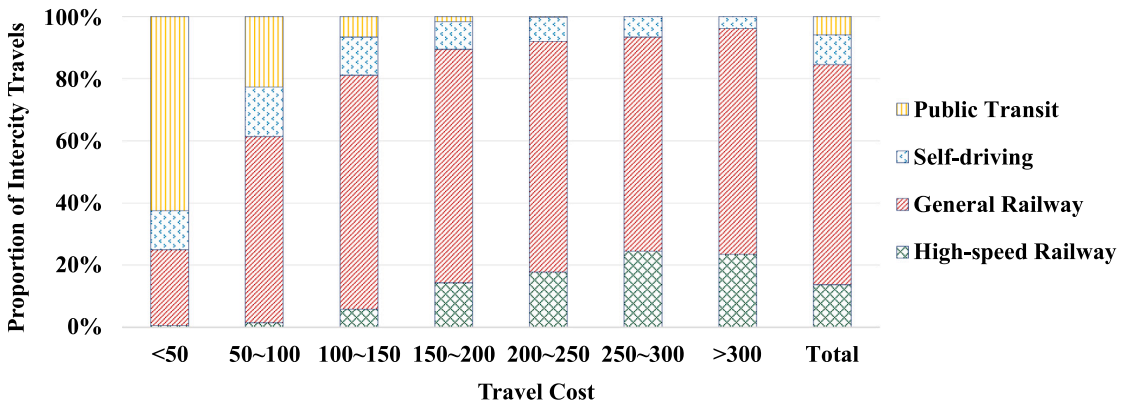


Fig. 7. Proportion of trips for which each intercity travel mode is the least costly based on our rule-constrained multi-modal route planning algorithm.

2013; Levinson and Kumar, 1994). This replicates the gravity-based measure, particularly when a fine-grained threshold setting is applied (Hansen, 1959).

$$A_{i,w_\tau} = \sum_{\tau \in T} (A_{i,\tau_n} - A_{i,\tau_{n-1}}) f(\tau_n) \tag{8}$$

subject to: τ that is sorted in an ascending order in set T

where:

A_{i,w_τ}: cost-weighted accessibility of county i;

τ_n: the n_{th} cost threshold in the threshold set T;

$f(\tau_n)$: travel cost decay function.

5. Results

5.1. Route planning statistics

The least cost routes between any two counties ($90 \times 89 = 8010$ in total) in the Guanzhong Plain Urban Agglomeration are found based on our proposed rule-constrained multi-modal route planning algorithm. Fig. 7 summarizes the share of these routes for which each intercity travel mode is selected, including driving and taking the HSR, GR, and public transit. Note that the highest priority is granted to the HSR mode in the counting, which means, if any of the travel segments between an OD pair includes HSR, the entire trip is categorized as an HSR trip. Trips including the GR (but excluding those involving any HSR segments) are counted as GR trips. Automobile trips are identified only when driving the sole mode used throughout the entire journey.

The results show that for 15% of the OD pairs, HSR has the lowest travel cost according to Eq. (5). Such a proportion is not as high as expected regarding its functional position, the backbone of the railway system, as well as its advantages on travel speed, but not quite surprising given the fact that only 28.9% of counties (26 out of 90) have HSR stations located within their geographical boundaries. Among these routes, driving is the dominant mode that connects the HSR stations (67.7%), mainly because most of the HSR stations reside in the periphery served poorly by public transit.

The GR provides the strongest supports for the intercity travel in the urban agglomeration. As shown in Fig. 7, more than half of the OD pairs use it as the primary intercity travel mode when optimizing the time plus monetary cost, in particular for the medium to long distance travel. This is acceptable given its features of high coverage rate and low fare level. The same as the HSR scenario, driving is still the dominant transfer mode for taking the GR, but its share is lower (55.9%) due to more center-located GR stations.

In addition, we cannot ignore that public transit serves travel for many adjacent counties, see Fig. 7 that the usage of public transit accounts for more than 60% of total for the OD pairs having a travel cost less than 50 CNY. This highlights the advantages of public transit regarding its lower fare. But a sharp decrease of its share can be recognized when the travel cost climbs, primarily because of the time wasted on circuitous routes and numerous transfers.

Driving comprises the rest of the intercity travel, most of which are shorter to medium distance trips.

These findings are further corroborated by manually tagged data from Amap, a mapping service equivalent to Google Maps in China. We randomly selected 100 OD pairs in each cost category across the Guanzhong Plain Urban Agglomeration and identified the lowest cost route for each by comparing Amap's recommended routes for HSR, GR, driving, and public transit, where available. The resulting proportions confirm trends similar to those observed in our study. More details could be found in Appendix B.

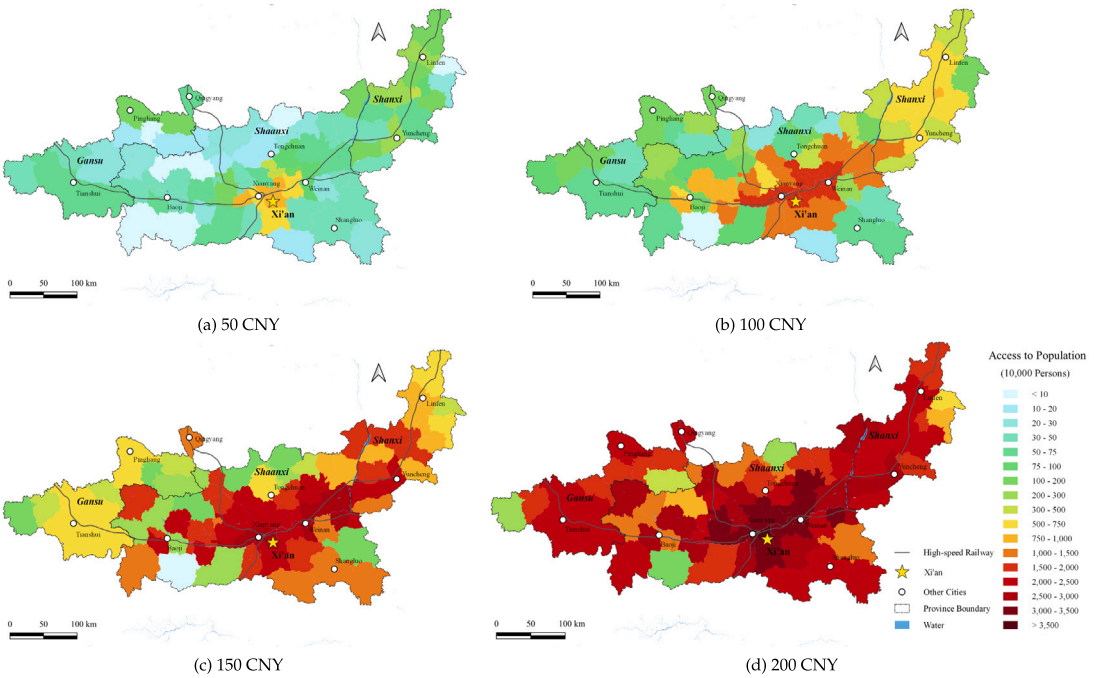


Fig. 8. Intercity accessibility measurements in different cost thresholds based on the multi-modal, multi-scalar networking approach.

5.2. Intercity access

Access to population based on the multi-modal, multi-scalar network is measured in different cost thresholds for the Guanzhong Plain Urban Agglomeration. The population-weighted average annual disposable income of the region, set at 15.31 CNY/hr, serves as the VoTT to integrate travel time and monetary costs. This value represents about 47.1% of the annual wage in the Xi'an metropolitan area (Statistics Bureau of Shanxi Province, 2022), aligning with guidelines from previous studies that recommend using 40% to 50% of wages to estimate VoTT (Hensher, 2019).

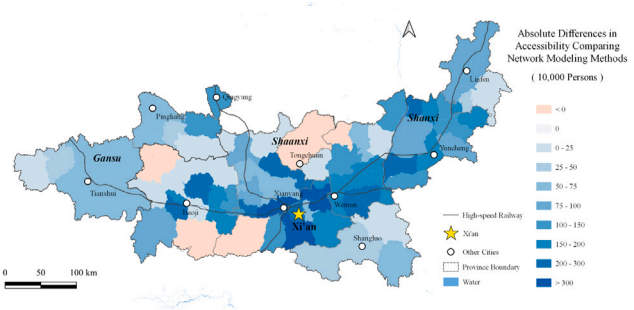
As shown in Fig. 8, expected general patterns of its spatial distribution can be found that the core city of the study area, Xi'an, experiences a higher level of accessibility, given the fact that both population opportunities and network densities, including both intercity transport networks as well as their feeder modes, are higher there. Its adjacent cities, like Xianyang and Weinan, also enjoy a higher accessibility. But the rest of the study area undertakes a relative lower accessibility, particularly when a lower cost threshold is applied, e.g., 50 CNY (Fig. 8(a)).

The cost threshold affects the accessibility positively that the warm-colored area expands sharply if a higher cost threshold is granted. As Fig. 8(d) shows, most of the counties and urban centers can reach more than 10 million people in 200 CNY, except those located far away from the HSR network, as the green-colored counties highlight. This implies that proximity to the HSR is a significant factor affecting intercity accessibility.

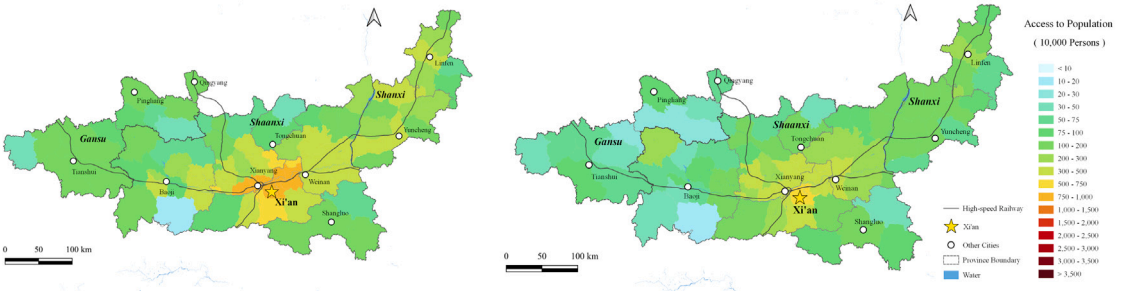
A direct comparison of the results between our multi-modal, multi-scalar networking approach and the traditional door-to-door approach[6] is shown in Fig. 9. In this comparison, the cost-weighted average accessibility are applied, as defined in Eq. (8), with decay functions calibrated using the Baidu migration data. The calibration process is detailed in Appendix C.

The figures demonstrate that the traditional door-to-door approach significantly underestimates intercity accessibility, with population-weighted averages shown as 424 (multi-modal networking) vs. 268 (door-to-door) in units of 10,000 persons. This underestimation primarily results from a segmented routing strategy, from origin to station, station to station, and finally station to destination, while neglecting additional connecting options, such as public transit. Furthermore, although the accessibility metrics from the two approaches show a high correlation (0.93), the impact of the door-to-door method is not uniformly distributed across all counties in the region. As shown in Fig. 9(a), it tends to overestimate accessibility in counties farther from the HSR lines due to its exclusive reliance on driving, so ignoring other transport modes as feasible alternatives.

[6] The door-to-door approach defines the "with-HSR" scenario as the option with the lower travel cost between directly driving from origin to destination and a combined route that includes driving to the nearest HSR station, using the HSR, and then driving from the station to the destination. A more detailed discussion of the door-to-door approach is available in Luo and Zhao (2021).



(a) Absolute differences in accessibility measurements (multi-modal networking minus door-to-door approach)



(b) Multi-modal networking approach (c) Traditional door-to-door approach

Fig. 9. Comparison of intercity accessibility measurements: multi-modal networking approach vs. traditional door-to-door approach.

5.3. Contribution of high-speed railways

To evaluate the contribution of the HSR to intercity travel, access to population is also measured based on the sub-multi-modal network, which excludes the HSR lines from the complete network. The absolute changes are visualized in Fig. 10, compared to the "with HSR" scenario (Fig. 8), where darker colors represent a greater impact of the HSR.

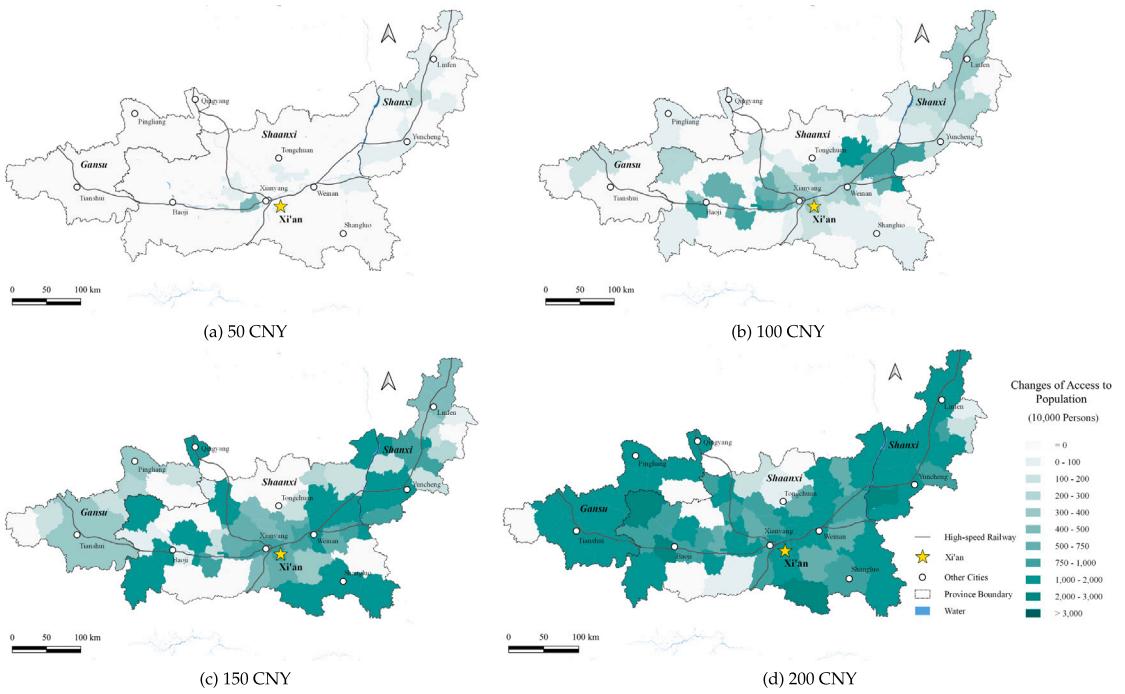
As the figures show, a positive effect of the HSR can be observed in any given cost thresholds, which demonstrates that the construction and operation of the HSR lines do benefit the intercity travel in terms of accessibility. Specifically, the HSR affects their nearby counties more significantly if a smaller cost threshold is considered, like Fig. 10(a) or Fig. 10(b) shows, that the green-colored areas are basically aligned along the network. But when the cost threshold increases, the influences are spatially spread out that, in 200 CNY, many marginal counties, regarding to Xi'an, can also earn accessibility benefits from the HSR. The whole story is expected as the HSR shows more advantages over other modes, for medium-to-long distance travel in particular, due to its higher speed than driving or taking urban transit. However, for shorter travel, e.g. from Xi'an to Xianyang, driving and urban transit can be a stronger competitor due to easier connections, so limits the spatial effects of the HSR. We consider driving and urban transit easier to connect as the former allows door-to-door travel without any further transfers, while urban transit has a denser network and shorter departure interval compared to the HSR.

A comparison between our multi-modal, multi-scalar networking approach and the traditional door-to-door approach in evaluating the impact of the HSR network on accessibility is presented in Fig. 11. As with accessibility measurements, the door-to-door approach tends to underestimate the HSR's impact, with the extent of this underestimation varying across the region, as shown in Fig. 11(a).

Notably, the correlation of accessibility improvement from the HSR network using these two approaches is only 0.68, compared to a correlation of 0.93 for accessibility itself. It suggests that the effects of networking methods are intensified in before-and-after analyses that compare changes rather than just evaluating static accessibility. These findings highlight the importance of accurate configuration of the network, especially with current trends towards integrated development, even when assessing the effectiveness of single-modal projects.

5.4. Sensitivity of Value of Travel Time (VoTT)

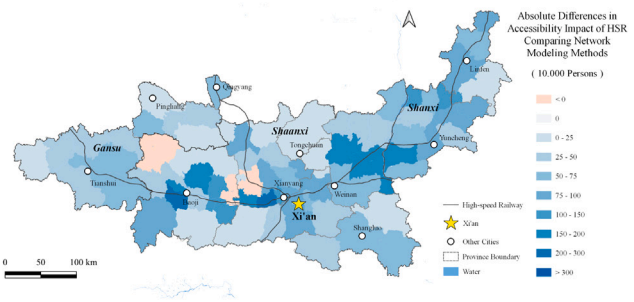
As illustrated in the definition of the cumulative opportunity measure (Section 4.3), the population-weighted average annual disposable income of the study area (15.31 CNY/hr) is considered as the baseline VoTT when combining the time and expense of



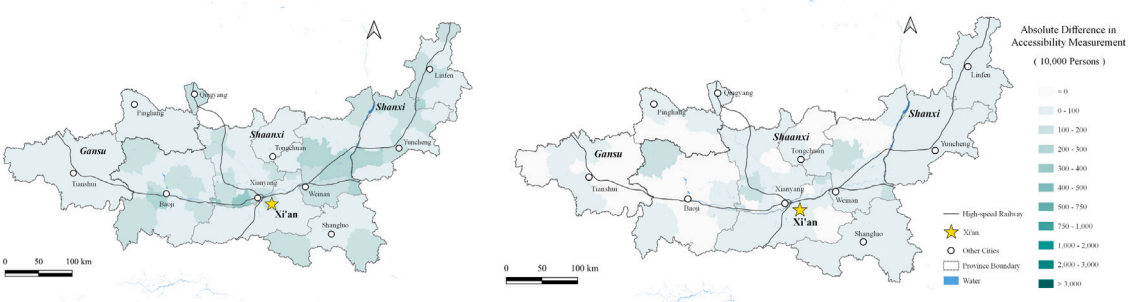
(a) 50 CNY (b) 100 CNY

(c) 150 CNY (d) 200 CNY

Fig. 10. Changes in intercity accessibility at different cost thresholds with and without the high-speed railway network based on the multi-modal, multi-scalar networking approach.



(a) Absolute differences in accessibility impact measurements (multi-modal networking minus door-to-door approach)



(b) Multi-modal networking approach (c) Traditional door-to-door approach

Fig. 11. Comparison of accessibility impact evaluations: multi-modal networking approach vs. traditional door-to-door approach.

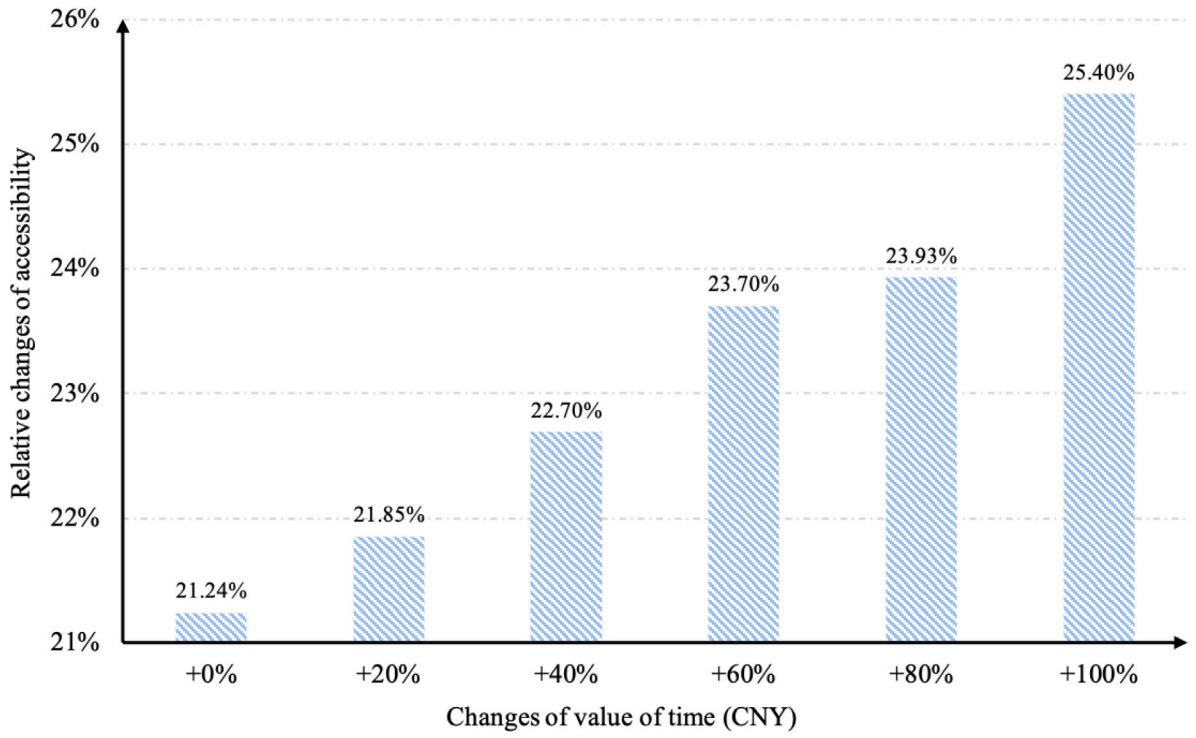


Fig. 12. Relative changes in population-weighted average access to population with and without high-speed rail network, based on the multi-modal, multi-scalar networking approach, accounting for increased value of travel time.

the intercity travel. A sensitivity analysis is conducted setting the VoTT varying from +0% to +100% to, at first, verify if the general conclusions on the HSR impacts remain unchanged when applying different settings of the VoTT, and, at second, to simulate the future spatial effect of the HSR with the process of economic development.

The spatial distributions of accessibility and its impact of the HSR network show the same patterns, as discussed in the corresponding subsections above, despite variations in the VoTT settings.

Fig. 12 further illustrates their relative changes in population-weighted average accessibility with-and-without the HSR network.

As we expect, the relative change in accessibility increases with a higher VoTT. This trend suggests that the spatial impacts of the HSR would increase with the pace of the economic development, considering the advantages of the HSR on time-savings over the other alternatives. Further expansions of the HSR network are expected by proponents to stimulate regional economic growth (Blanquart and Koning, 2017; Li et al., 2020; Wu and Levinson, 2024). This forms a positive feedback system, but whether the HSR construction leads economic growth, or the other way around, is the another research question (Li et al., 2021), to which our best estimates indicate that HSR leads per capita GDP growth, but not vice versa (Wu and Levinson, 2024). Future studies can continue to explore the evolutionary interactions between the changes in travel demand and economic productivity due to network expansion.

An additional sensitivity analysis is also conducted to explore how variations in the VoTT across different counties and among individuals within those counties impact the accessibility changes due to the HSR. To do so, we assume a hypothesized VoTT distribution based on city-level annual average disposable income data, and incorporate probabilistic fluctuations into our fixed VoTT setting. Multiple accessibility evaluations are then performed with these varying VoTT values; however, only minor variations in the results can be observed. This outcome verifies the reliability of using a fixed VoTT setting in macroscopic accessibility evaluations. Detailed discussions on the methodology and results are provided in Appendix D.

6. Conclusion

The importance of multi-modal transport planning has been recognized in recent years, but has not been thoroughly considered in practical applications. This paper proposes a multi-modal intercity accessibility framework that employs a multi-modal, multi-scalar networking approach to simulate the connections and transfers between various intercity and intra-city networks, and applies it to explore the impacts of the HSR on accessibility.

Specifically, the multi-modal, multi-scalar networking approach integrates multiple layers of transport networks, from the perspectives of both intercity and intra-city travel. It then coordinates the actual route and mode choice logic with the theoretical optimal route planning strategy to mitigate potential conflicts that may render the shortest path infeasible within the multi-modal

network. The route planning algorithm finds the least general cost routes for every OD pair in the study area and returns the corresponding mode(s) as well as the travel time and monetary cost en-route. These outputs serve as precise measurements of intercity travel impedance, providing essential inputs for accessibility calculations, which allows for a better illustration of the competitive and cooperative dynamics among different transport modes within the context of multi-modal integrated development.

The Guanzhong Plain Urban Agglomeration has been selected as the study area for a proof-of-concept evaluation in this research. It serves two primary purposes: first, to verify the practical applicability of the proposed multi-modal accessibility framework in real-world settings, and second, to analyze the impact of the HSR network on accessibility within the Guanzhong Plain Urban Agglomeration, guiding future development efforts within an integrated multi-modal network.

The results indicate that Xi'an, the core city of the study area, has better access to population centers, where both opportunities and transport interchanges are clustered. Additionally, counties located near the HSR lines experience higher intercity accessibility, highlighting the HSR's role in enhancing connectivity within urban agglomerations. The accessibility impact of the HSR within the multi-modal, multi-scalar network further corroborates the finding that the HSR does improve intercity accessibility, with significant increments observed in nearby counties. When a higher cost threshold is applied, noticeable positive impacts are also recognized in more distant counties, which may require multiple transfers or longer drives to access the HSR, but illustrating the radiation effect of the HSR network. However, it is noteworthy that the current proportion of intercity travel using the HSR remains low from a route planning perspective.

These results, while expected, differ from those obtained through the traditional door-to-door approaches that simply use auto as the connecting mode and segmentally identify the shortest paths for measuring travel impedance without capturing the dynamic transfer processes at transport hubs. In general scenarios focusing on a single transport mode investment, such as HSR line expansion, a more accurate accessibility evaluation could prevent biased assessments of proposals (Cui and Levinson, 2018), e.g., underestimating accessibility benefits by overlooking the complementary roles of other modes that enable connections to the HSR network or overestimating HSR's contributions by failing to account for competition among alternative transport options.

Furthermore, this multi-modal, multi-scalar accessibility framework is particularly beneficial for investments involving multiple transportation modes. For instance, the Xi'an East station, a multi-modal transport hub scheduled to open in early 2026, is designed to integrate HSR, GR, metro services (three metro lines), and a bus terminal. Our approach effectively addresses the challenges of modeling connection efficiency among different modes, where traditional accessibility measures often fall short. Decision-makers can identify optimal configurations, e.g., overall locations or detailed layouts, by evaluating and comparing potential gains in accessibility and possible losses in social equity across various proposals, from both intracity and intercity perspectives.

The methodology, including both the networking approach and the accessibility framework, can be extended and applied to other regions if relevant network, travel impedance metrics, and opportunity data are available. Note that generalized travel cost, which encompasses travel time and monetary expenses, serves as the impedance for accessibility calculations. These components are key considerations in route and mode selection, thus better reflecting the interrelationship among various transportation modes. However, it requires careful configuration of the VoTT to effectively integrate these two components.

In accessibility evaluations, deterministic values derived from personal income are generally preferred when configuring the VoTT, since decision-makers typically seek definitive conclusions rather than uncertain variations. Additionally, averaged results from multiple iterations with probabilistic fluctuations around the fixed VoTT often yield results that closely align with the fixed value itself. However, for studies aiming to better model individual preferences within the context of multi-modal networks, such as route or mode choice modeling (Axhausen et al., 2016; Zhang and Levinson, 2004; Cui and Levinson, 2021) or person-based accessibility analysis (Dianin et al., 2024; Huang, 2020), rather than adopting a macroscopic planning perspective, a probabilistic method of VoTT setting may be more effective in capturing the impact of VoTT variations on individual travel behaviors.

In addition, the route planning strategies in our study ignores secondary factors that may influence travelers' decisions. While the shortest travel time path and the lowest travel cost path are commonly used to estimate the OD travel impedance in accessibility analysis, consistent with the approach taken in this analysis and numerous previous studies, more complete route planning rules that consider complex influencing factors could explain more of the variance in individual preferences. Future studies on microscopic travel behavior should explore these aspects further.

Note also that the accessibility measure used in this study, while consistent with many previous analyses, does not account for the service frequency of transport modes when assessing intercity accessibility and the impact of the HSR network. For intercity travel segments, the waiting time at the intercity transport hubs on the origin side is hypothesized as a fixed value based on existing literature. This approach is particularly suitable for scenarios where travel is pre-planned, as it assumes travelers arrive at transport hubs approximately 30 min before departure, thereby diminishing the importance of service frequency. The inherent flexibility of connecting modes, such as public transit, driving, and ride-hailing services, can significantly influence travelers' mode choices and, thereby, overall accessibility. For example, if a traveler intends to catch a HSR train scheduled for 6:00 am, driving or using ride-hailing services might be their only viable options, as public transit services may not yet be available prior to that hour. While the waiting time associated with transit schedules becomes less critical in the context of intercity accessibility analysis, particularly in cities like Xi'an, where public transport operates at exceptionally high frequency over most of the day. Therefore, for future studies, especially those aimed at assessing daily intercity accessibility that often involves spontaneous or routine commuting, this multi-modal accessibility framework should be extended to incorporate the operational frequency of transport modes into accessibility measures.

CRediT authorship contribution statement

Zhe Dai: Validation, Methodology, Funding acquisition, Conceptualization. **Mengying Cui:** Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Kong Li:** Validation, Software, Investigation. **Lijie Yu:** Resources, Data curation. **David Levinson:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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Appendix A. Rule-constrained multi-modal route planning

The rule-constrained multi-modal route planning algorithm proposed in this study addresses the shortcomings of existing shortest path searching strategies in multi-modal networks. These traditional methods often yield infeasible routes that, while optimal from a cost perspective, lead to implausible cross-network transfers. Our algorithm consists of three main steps:

- Step 1: constructing a route set that includes one basic route and two subsets based on the multi-modal network (see Fig. 13);
- Step 2: establishing rules to avoid potential conflicts that could render shortest path searching theoretically infeasible within the multi-modal network (see Fig. 14(a) and (b));
- Step 3: implementing rule-constraints into the path searching algorithm to create the rule-constrained multi-modal route planning algorithm for identifying plausible optimal cross-network routes between specific OD pairs (see Fig. 14(c) and (d)).

The pseudo code of the algorithm can be found in Algorithm 1.

Step 1: Constructing the multi-modal route sets

Fig. 13 illustrates the process of constructing multi-modal route sets for a given OD pair.

- First, we constructed three distinct route sets based on the multi-modal network (see Fig. 13(a)) for railway, public transit, and driving, as shown in Fig. 13(b), (c), and (d), respectively.
- Subsequently, we simplified the three route sets using the principle of minimum cost, resulting in simplified graphs for each mode of travel, as illustrated in Fig. 13(e), (f), and (g).
- Finally, by integrating the three modes of travel and combining them in pairs, we developed the final multi-modal route sets, which include a basic route set (see Fig. 13(i)) and two route subsets (see Fig. 13(h) and (g)).

Step 2: Establishing rules to avoid potential conflicts

When traditional shortest path searching methods are applied to multi-modal transport networks, such as the one depicted in Fig. 14(a), the resulting configurations shown in Fig. 14(b) reveal an instance of transfers involving three modes of transportation, occurring up to five times, which are considered infeasible. To address this issue, we propose certain rules to constrain the shortest path searching within the multi-modal network:

- Rule 1: When optimal paths involve segments of HSR or GR, only one transfer between different modes of transportation, either for access to or egress from an intercity transport hub, should be permitted;
- Rule 2: When optimal paths involve segments of HSR or GR, a maximum of two transfers between different railway lines should be allowed;
- Rule 3: When optimal paths involve segments of HSR or GR, no driving or public transit modes should be located between two railway segments, except in cases of inner-city railway transfers, where one city includes more than one intercity transport hub. The number of transfers for the connecting mode should adhere to Rule 1;
- Rule 4: When optimal paths do not involve segments of HSR or GR, a maximum of two transfers between different modes of transportation should be permitted.

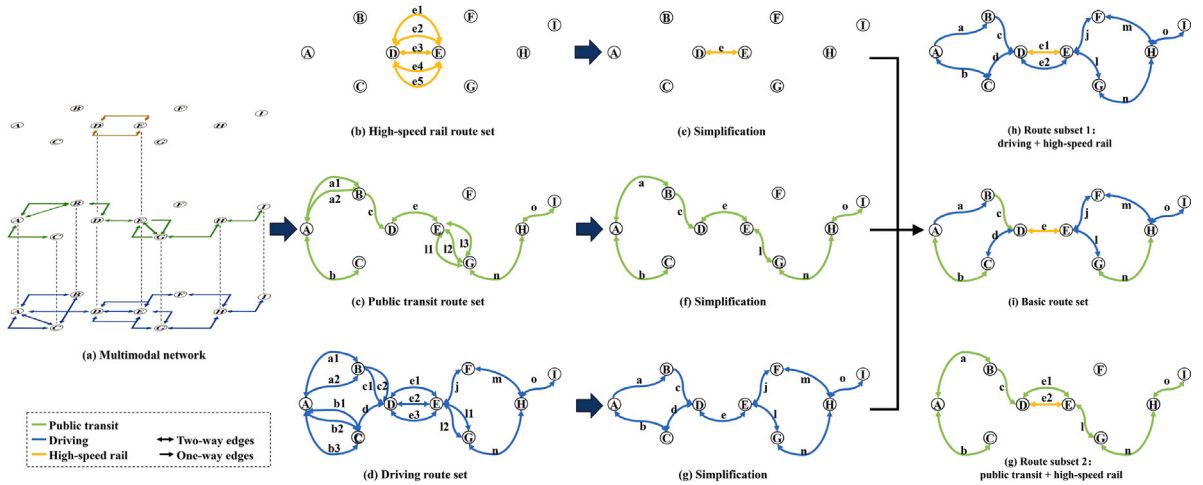


Fig. 13. An illustration of multi-modal route set construction, where uppercase letters "A" to "H" represent nodes, e.g., HSR stations, while lowercase letters "a" to "n" denote edges that represent paths for specific transportation modes.

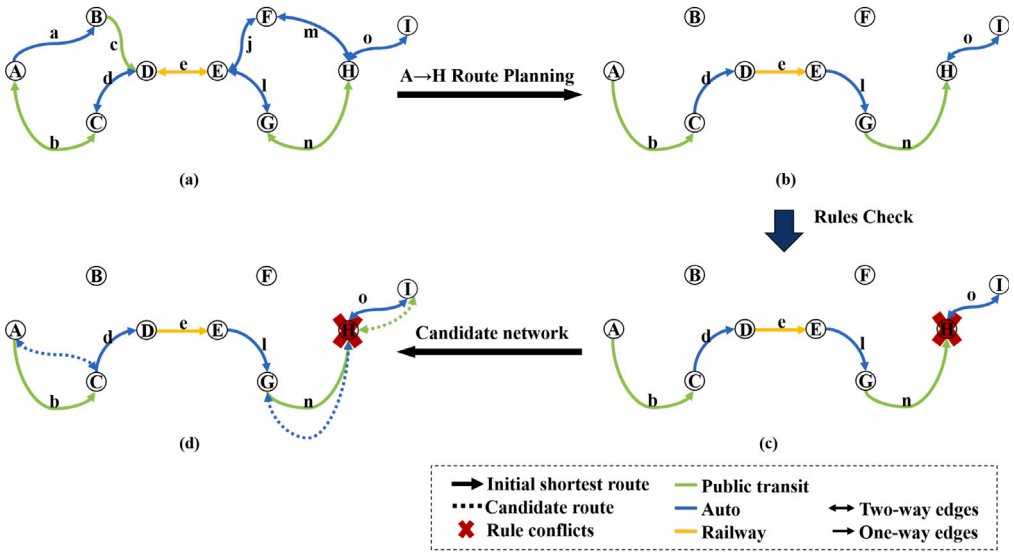


Fig. 14. An illustration of the rule constraints, where the uppercase letters "A" to "H" represent nodes, e.g., HSR stations, while lowercase letters "a" to "n" denote edges that represent paths for specific transportation modes. (a) shows the basic route set of the multi-modal network with nine nodes; (b) displays the identified shortest path from the basic route set; (c) presents a case where a conflict against the rule constraints is observed; (d) indicates the candidate route that adheres to the rules while also minimizing the cost.

Step 3: Implementing the rule constrains to the path searching algorithm

In practical applications, we propose a rule-constrained path planning algorithm, detailed in Algorithm 1, which consists of three main steps:

- First, we use a graph traversal algorithm to identify the optimal path within the basic route set.
- Next, we perform a rule check on the identified shortest path. If the path complies with the established rules, it is selected as the optimal path. If not, as illustrated in Fig. 14(c), we will conduct a re-evaluation by combining and assessing the shortest paths obtained from graph traversal searches in the two route subsets, as the candidate routes shown in Fig. 14(d).
- Ultimately, the algorithm returns a path that complies with the established rules and minimizes the total travel weight as the final output.

Algorithm 1: Rule-constrained multi-modal route planning

Input : Multi-modal graph G, Origin point o, Destination point d
Output: The optimal route R from o to d

Function:
  Step 1: Multi-modal route sets construction
  Build the multi-modal route set G_b and subsets G_{s_i} based on G
  go to Step2
  Step 2: Route Searching
  R_b = RS(G_b, o, d) // Find the shortest path R_b from o to d on G_b with a graph traversal algorithm
  go to Step3
  Step 3: Rule Checking
  if R_b *fit the rules* **then**
    $R = R_b$
    go to Step 5
  else
    go to Step 4
  end
  Step 4: Rule-constrained Optimal Route Searching
  R_i = RS(G_{s_i}, o, d)
  if $R_p \cap R_q$ ($p \neq q$) *is not* $\emptyset$ **then**
    initialization: R_{pq} // overlapping segment
    R_{p*} // none-overlapping segment in R_p
    R_{q*} // none-overlapping segment in R_q
    R_{pq*} = min(R_{p*}, R_{q*}) // Select the path with the minimum travel impedance between R_{p*} and R_{q*} for R_{pq*}
    $R = R_{pq} \cup R_{pq*}$
    go to Step 5
  else
    R = min(R_p, R_q) // Select the path with the minimum travel impedance between R_p and R_q for R
    go to Step 5
  end
  Step 5: Return Optimal Route R
  return R
end

Appendix B. Validation of route planning statistics

For verification of our rule-constrained multi-modal route planning algorithm, we randomly selected 100 OD pairs in the Guanzhong Plain Urban Agglomeration across different cost categories. We manually collected their recommended routes for HSR, GR, driving, and public transit, where available, and tagged the lowest cost option as the optimal choice. The proportions of each mode within each cost category, following the categorization in Fig. 7, was counted and the results are displayed in Fig. 15.

The analysis confirms similar trends to those observed with our route planning algorithm, with GR providing substantial support for intercity travel within the urban agglomeration. Public transit is particularly effective for travel between adjacent counties, while driving predominantly covers shorter to medium distance trips. The RMSE between the proportions of each mode by cost category, comparing manually tagged data with our algorithm, is approximately 11.58%.

Note that Amap does not provide optimal routes in the context of a multi-modal network. Users must select their preferred mode of transport, after which Amap returns the optimal route for that mode. In the case of intercity travel, both HSR and GR routes can be obtained through the public transit option, which may also include urban transit routes if they are available. However, Amap considers only driving as the connecting mode to intercity transport hubs when calculating travel impedance factors like time and expenses.

Appendix C. Decay function calibration

The decay functions are calibrated for each VoTT setting, ranging from +0% to +100%. The calibration results are displayed in Fig. 16.

To do so, we calculate the percentage of the daily migration flow in each of the travel cost categories, e.g., 0–50 CNY, 50–100 CNY, and so on, where the migration flow between two cities refers to their yearly average migration scale index divided by the

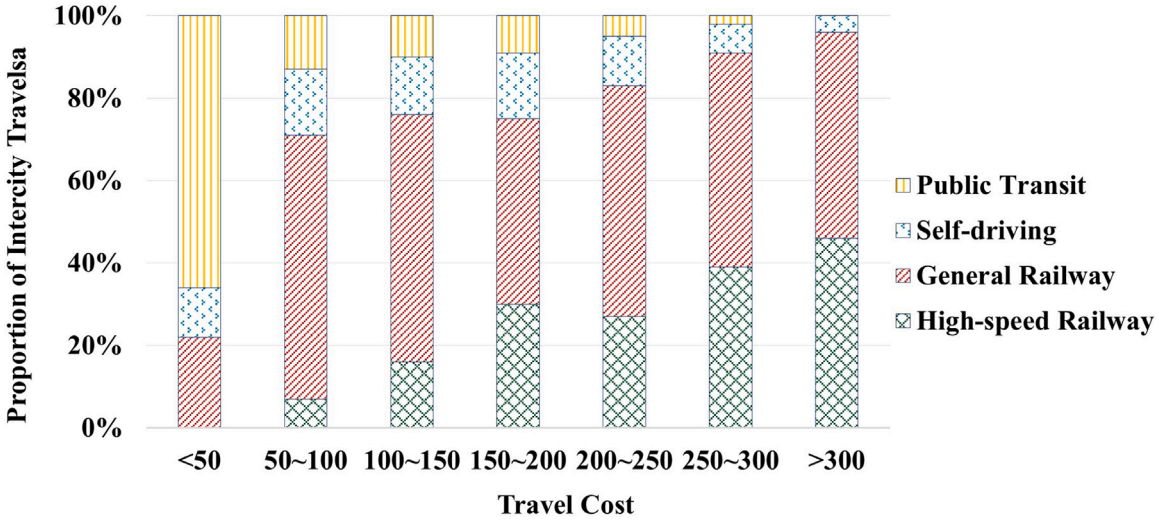


Fig. 15. Proportion of trips for which each intercity travel mode is the least costly based on manually tagging data from Amap.

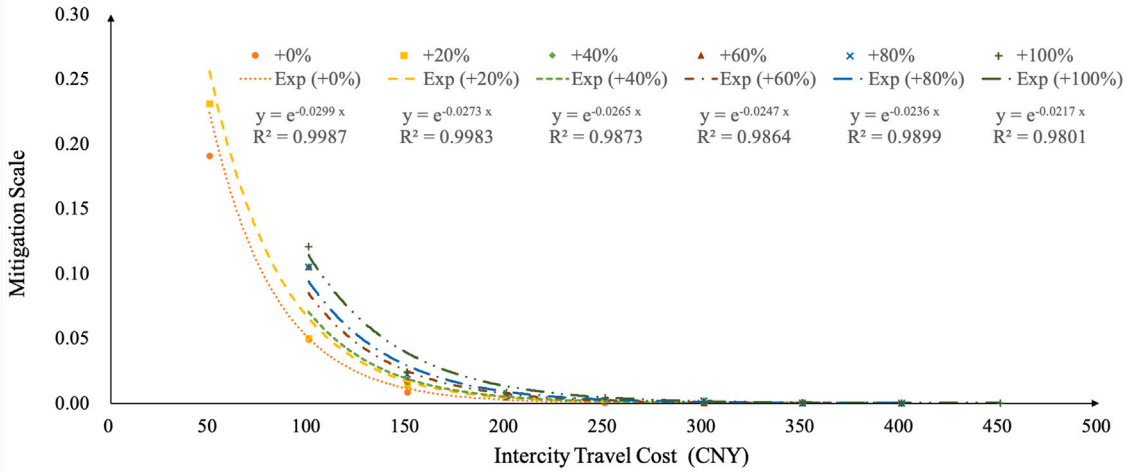


Fig. 16. Travel cost decay functions for intercity travel with the value of travel time increases from 0% to 100%.

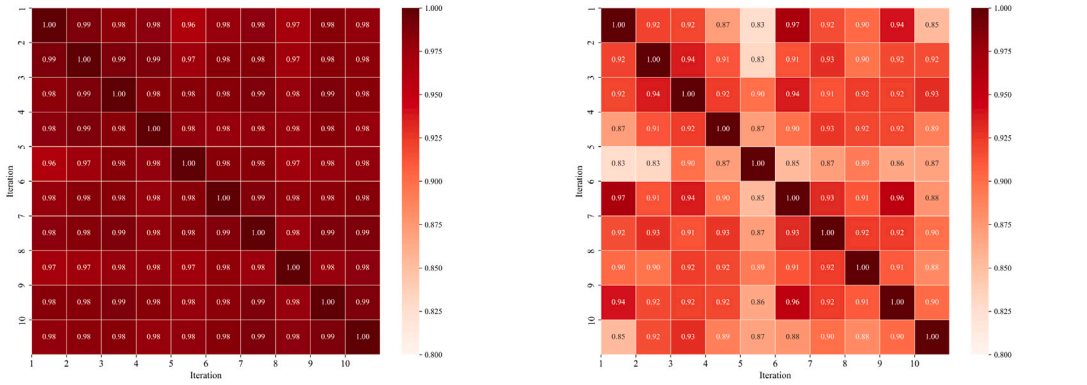
population in total, and their travel cost is measured based on our proposed network. This data processing here aims to eliminate the effects of the city scales on the interaction strength, and only capture the part due to the extent of the spatial separation.

The least squares curve fitting algorithm is used to calibrate the decay function. It selects the exponential format of function because, first, it gives a better fit than the power and natural log functional forms, and, second, previous studies used the exponential form to explain the decay of urban travel (Cui and Levinson, 2020b; Anderson et al., 2013). Note that the same intercept (intercept = 1) is set for all VoTT settings for a more intuitive comparison.

Appendix D. Sensitivity analysis of VoTT regarding probabilistic fluctuations

This sensitivity analysis assesses the reliability of deterministic VoTT settings in evaluating intercity accessibility from a macroscopic planning perspective. For this analysis, we model the VoTT as a normal distribution, with a mean of 15.31 and a standard deviation of 3.92. These parameters are set such that the mean is equal to the population-weighted average annual disposable income, with the standard deviation also adjusted by population weights. We recognize that income typically exhibits a right-skewed distribution. However, employing population weights helps mitigate the impact of the skewed tail, making the normal approximation more reasonable. Future research focused on travel behavior or the like should consider incorporating a more accurate distribution model for VoTT if detailed data is available.

A probabilistic approach is implemented where a different VoTT, drawn from the distribution described above, is assigned to each OD pair to identify the optimal path, measure the cost impedance, and calculate accessibility. To verify the robustness of the



(a) Accessibility metrics

(b) Accessibility changes due to the HSR network

Fig. 17. Correlation matrices for accessibility metrics and high-speed railway (HSR) network impact across ten iterations with value of travel time fluctuations.

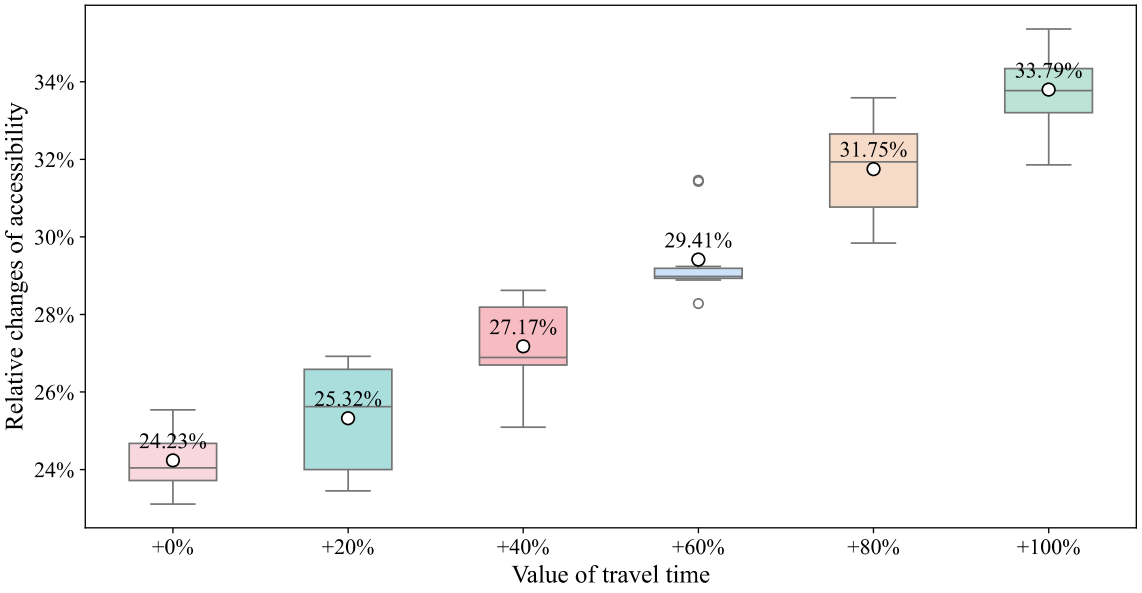


Fig. 18. Relative changes in population-weighted average access to population with and without high-speed rail network, based on the multi-modal, multi-scalar networking approach, accounting for increased value of travel time and its fluctuations.

analysis, ten iterations are conducted, and the correlations of accessibility metrics and the improvements from the HSR network across these iterations are detailed in Fig. 17.

The results demonstrate that accessibility metrics across different iterations are highly correlated, indicating that fluctuations in the VoTT do not significantly impact accessibility evaluations on such a large geographical scale. While the accessibility changes due to the HSR network shows more significant variation with VoTT fluctuations, the lowest correlation between any two iterations is still above 0.8. Moreover, averaging results from these 10 iterations effectively reduces the impacts of VoTT fluctuation, with correlations reaching 0.99 for accessibility metrics and 0.97 for accessibility changes when compared to the fixed VoTT setting. We expect that the effect of VoTT fluctuations will be further reduced with more iterations, potentially eliminated entirely with an infinite number of iterations.

In addition, we also applied fluctuations to each base VoTT setting, varying from 0% to +100%, to simulate the impact of the HSR network alongside economic development. The relative changes in population-weighted average accessibility to population, with and without the HSR integrated in the multi-modal network, are illustrated in Fig. 18. As it shows, while fluctuations can be observed, the trends remain the same as Fig. 12, indicating that the spatial impacts of the HSR network increase with economic development.

These findings confirm that our fixed VoTT setting is reliable for planning practice, producing outcomes consistent with those considering VoTT fluctuations.

Data availability

Data will be made available on request.

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